

Markov Chain Intro

Monte Carlo Statistical Methods

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1 Level 0: Core Foundations of Discrete-Time Markov Chains

1.1 Definition of a Markov Chain

A discrete-time Markov chain is a stochastic process

$$X_0, X_1, X_2, \dots$$

that satisfies the Markov property

$$\mathbb{P}(X_{n+1} = y \mid X_0, X_1, \dots, X_n) = \mathbb{P}(X_{n+1} = y \mid X_n).$$

Intuition. The next step depends only on where you are now. The past does not influence the future once the present state is known.

1.2 Types of State Spaces

- Finite state space:

$$\mathcal{X} = \{1, 2, \dots, m\}.$$

- Countably infinite:

$$\mathcal{X} = \mathbb{N}, \mathbb{Z}, \text{ etc.}$$

- Continuous or uncountable:

$$\mathcal{X} \subseteq \mathbb{R}^d.$$

1.3 Transition Rule

1.3.1 Finite and Countable State Space

The transition probabilities are

$$p_{ij} = \mathbb{P}(X_{n+1} = j \mid X_n = i), \quad p_{ij} \geq 0, \quad \sum_j p_{ij} = 1.$$

The transition matrix is

$$P = \begin{pmatrix} p_{11} & p_{12} & \cdots & p_{1m} \\ p_{21} & p_{22} & \cdots & p_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ p_{m1} & p_{m2} & \cdots & p_{mm} \end{pmatrix}.$$

Each row i is a probability distribution over next states.

1.3.2 Continuous State Space

When $\mathcal{X} \subseteq \mathbb{R}^d$, we have a transition density

$$T(x, y) = p(y \mid x), \quad T(x, y) \geq 0, \quad \int_{\mathcal{X}} T(x, y) dy = 1.$$

Intuition: At every point x , there is a full density over all possible next states y . This plays the exact same role as the row of a matrix in the discrete case.

2 Evolution of Distributions

Let μ_n denote the distribution of X_n . For finite $m = |\mathcal{X}|$, write it as a row vector

$$\mu_n = [\mu_n(1), \mu_n(2), \dots, \mu_n(m)].$$

2.1 Discrete State Evolution

The evolution rule is

$$\mu_{n+1}(j) = \sum_{i=1}^m \mu_n(i) p_{ij}.$$

In matrix form:

$$\mu_{n+1} = \mu_n P.$$

2.1.1 Expanded Example

Let

$$\mu_n = [a_1, a_2, \dots, a_m]$$

and

$$P = \begin{pmatrix} p_{11} & p_{12} & \cdots & p_{1m} \\ p_{21} & p_{22} & \cdots & p_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ p_{m1} & p_{m2} & \cdots & p_{mm} \end{pmatrix}.$$

Then

$$\begin{aligned} \mu_{n+1}(1) &= a_1 p_{11} + a_2 p_{21} + \cdots + a_m p_{m1}, \\ \mu_{n+1}(2) &= a_1 p_{12} + a_2 p_{22} + \cdots + a_m p_{m2}, \\ &\vdots \\ \mu_{n+1}(m) &= a_1 p_{1m} + a_2 p_{2m} + \cdots + a_m p_{mm}. \end{aligned}$$

Every component is a weighted sum of a column of P . Multiplying by P simply updates the distribution to the next step.

2.2 Continuous State Evolution

When densities exist:

$$\mu_{n+1}(y) = \int_{\mathcal{X}} \mu_n(x) T(x, y) dx.$$

Intuition: Replace the discrete summation by integration. Each x contributes density toward all possible y according to $T(x, y)$.

3 Chapman Kolmogorov Equations

3.1 Discrete Case

For all n, m ,

$$p_{ij}^{(n+m)} = \sum_k p_{ik}^{(n)} p_{kj}^{(m)}.$$

Matrix form:

$$P^{n+m} = P^n P^m.$$

3.2 Continuous Case

For all n, m ,

$$T^{n+m}(x, z) = \int_{\mathcal{X}} T^n(x, y) T^m(y, z) dy.$$

Same idea as discrete, with integrals instead of sums.

4 Operator View

Define a function $f : \mathcal{X} \rightarrow \mathbb{R}$. For a finite state space with m states, write

$$f = \begin{pmatrix} f(1) \\ f(2) \\ \vdots \\ f(m) \end{pmatrix}.$$

4.1 Discrete Case: Expanding the Operator

The operator P acts on f by

$$(Pf)(i) = \sum_{j=1}^m p_{ij} f(j) = \mathbb{E}[f(X_{n+1}) \mid X_n = i].$$

In vector form:

$$Pf = \begin{pmatrix} p_{11}f(1) + p_{12}f(2) + \cdots + p_{1m}f(m) \\ p_{21}f(1) + p_{22}f(2) + \cdots + p_{2m}f(m) \\ \vdots \\ p_{m1}f(1) + p_{m2}f(2) + \cdots + p_{mm}f(m) \end{pmatrix}.$$

Intuition: The i 'th component of Pf is the expected value of f one step ahead, conditioned on being in state i now.

4.2 Continuous Case: Density Kernel Version

For $x \in \mathcal{X}$,

$$(Tf)(x) = \int_{\mathcal{X}} T(x, y) f(y) dy = \mathbb{E}[f(X_{n+1}) \mid X_n = x].$$

Intuition is identical to the discrete case. Summations turn into integrals. Rows of a matrix become conditional densities.

5 Why Level 0 Matters

The ideas in this section form the backbone for all higher levels:

- Reachability and communication classes
- Recurrence and transience
- First return times and stationary distributions
- Convergence theorems and ergodicity
- MCMC kernels and detailed balance

They all start from understanding how a Markov chain transitions, how distributions evolve, and how operators act.